

significantly degraded. By adapting and changing the strategy, Thorp was able to consistently maintain extremely strong return/risk performance, whereas staying with the original system because it had worked so well at one time would have fared poorly.

What is the optimal size for any trade? Theoretically, this question has a precise answer: the Kelly criterion, which it can be mathematically demonstrated will yield a higher cumulative return over the long run than any other strategy for determining trading size. There is a rub, however. The Kelly criterion assumes that the probability of winning and the ratio of the amount won to the amount lost per wager are precisely known. Although this assumption is typically true for games of chance, in trading, the probability of winning is unknown and, at best, can only be estimated—often with a wide degree of error. If the Kelly criterion is used to determine the trade size, there is a steep penalty for overestimating the probability of winning (or the ratio of the average win to average loss). In fact, the negative impact of overestimating the correct trade size is twice as large as the negative impact of underestimating the correct trade size by the same amount. Therefore, if the precise probabilities of winning are not known—as is usually the case for trading applications—then the bet size should be significantly smaller than the full Kelly amount.

Also, even assuming the correct Kelly trade size was precisely known, the resulting equity stream would be far too volatile for most people's comfort level. The high volatility implicit in the Kelly criterion is not merely an aesthetic issue, but has important practical implications as well. The greater the volatility of the equity stream, the greater the chance a trader will abandon the approach on one of the drawdowns. Thorp recommends that using half Kelly is a better alternative (than full Kelly) for most people, even if the probability of winning (and the average win/average loss ratio) can be reasonably estimated. If the estimate for the probability of winning is subject to wide error, then only a very small fraction of Kelly should be risked on any trade. When Thorp was faced with such a situation (that is, when he traded a trend-following system), he estimates that his trade size was probably less than one-tenth of Kelly—such a small fraction of Kelly that he didn't even use the Kelly criterion to size trades.